Lecture 4: Series

Mathematical Analysis

1 Subsequence

Definition 1.1 (Subsequence). [def:subsequence] If a sequence $n : \mathbb{N}_1 \to \mathbb{N}_1$ is strictly increasing, then the sequence $b : \mathbb{N}_1 \to F = a(n(k))$ is called a *subsequence* of $a : \mathbb{N}_1 \to F$.

Theorem 1.1 (Subsequence Index Grows). [thm:subsequence-index-grows] *If $b : \mathbb{N}_1 \to F = a(n(k))$ is a subsequence of $a : \mathbb{N}_1 \to F$, then $\forall k \in \mathbb{N}_1, n(k) \geq k$.*

Proof. Since $n : \mathbb{N}_1 \to \mathbb{N}_1$ is strictly increasing, we have $n(1) < n(2) < n(3) < \cdots$.
Notice that $n(1) \geq 1$, since $n(1) \in \mathbb{N}_1$.
Also, if $n(k) \geq k$, then $n(k+1) > n(k) \geq k$, so $n(k+1) \geq k+1$.
By induction, we have $\forall k \in \mathbb{N}_1, n(k) \geq k$. □

Theorem 1.2 (Subsequence of a Subsequence). [thm:subsequence-of-subsequence] *A subsequence of a subsequence is still a subsequence of the original sequence.*

Remark 1.1 (Intuition: Picking Forward). [rem:subsequence-intuition] A subsequence can be understand as picking element from a sequence, however during picking we can only go forward, cannot go back to pick an element or pick an element twice

Theorem 1.3 (Limit of a Convergent Subsequence). [thm:convergent-subsequence-limit] *The limit of a subsequence is the same as the limit of the original sequence.*

Proof. **TODO** □

Corollary 1.1 (Diverges via Two Limits). [cor:divergence-via-subsequences] *If a sequence have two subsequences with different limits, then the sequence diverges.*

Proof. **TODO** □

2 Bolzano-Weierstrass Theorem

Theorem 2.1 (Bolzano-Weierstrass Theorem). [thm:bw] *Every bounded sequence $a : \mathbb{N}_1 \to F$ where F is a complete ordered field has a convergent subsequence.*

Proof. **TODO** □

3 Cauchy Criterion

Definition 3.1 (Cauchy Sequence). [def:cauchy-sequence] $a : \mathbb{N}_1 \to F$ is a *Cauchy sequence* if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N}_1,\ \forall n, m \in \mathbb{N}_1\ n > N \text{ and } m > N \implies d(a_n, a_m) < \varepsilon$$

Theorem 3.1 (Cauchy Criterion). [thm:cauchy-criterion] *If F is complete ordered field, then $a : \mathbb{N}_1 \to F$ is a Cauchy sequence if and only if it converges.*

Proof. **TODO** □

Theorem 3.2 (Cauchy Implies Bounded). [thm:cauchy-bounded] *Cauchy sequence are bounded.*

Proof. **TODO** □

4 Series

Definition 4.1 (Series Operator). [def:series-operator] Let F be a field. The *series operator*

$$\text{series} : (\mathbb{N}_1 \to F) \to (\mathbb{N}_1 \to F)$$

is defined recursively by

$$\text{series}(a)(1) := a(1), \qquad \text{series}(a)(n+1) := \text{series}(a)(n) + a(n+1).$$

Definition 4.2 (Series). [def:series] Let F be a field and $a : \mathbb{N}_1 \to F$. The *series of* a is the sequence

$$\text{series}(a) : \mathbb{N}_1 \to F,$$

obtained by applying the series operator to a. Its n-th term is the partial sum $\sum_1^n a$.

Definition 4.3 (Convergent Series). [def:convergent-series] Let F be a field and $a : \mathbb{N}_1 \to F$. The series of a is *convergent* if $\text{series}(a)$ is a convergent sequence.

Definition 4.4 (Infinite Summation). [def:infinite-series] Let F be a field and $a : \mathbb{N}_1 \to F$ with a convergent series. The *infinite summation* of a is

$$\sum a := \lim(\text{series}(a)).$$

Theorem 4.1 (Linearity of Infinite Summation). [thm:linearity-of-infinite-summation] *Let F be a field and $a, b : \mathbb{N}_1 \to F$ with convergent series. Then*

$$\sum(a+b) = \sum a + \sum b, \qquad \sum(\lambda a) = \lambda \sum a.$$

Theorem 4.2. *A series of a sequence $a : \mathbb{N}_1 \to F$ where F is a complete ordered field converges only if a converges to 0.*

Proof. **TODO**(Use cauchy criterion to prove this theorem) □

5 Comparison Test for Sequence

Lemma 5.1 (Series with Non-Negative Terms is Increasing). [lem:series-nonneg-increasing] *Let F be an ordered field and $a : \mathbb{N}_1 \to F$ with $a(n) \geq 0$ for all $n \in \mathbb{N}_1$. Then* $\text{series}(a)$ *is an increasing sequence.*

Lemma 5.2 (Series with Positive Terms is Strictly Increasing). [lem:series-positive-strictly-increasing] *Let F be an ordered field and $a : \mathbb{N}_1 \to F$ with $a(n) > 0$ for all $n \in \mathbb{N}_1$. Then* $\text{series}(a)$ *is a strictly increasing sequence.*

Corollary 5.1 (Comparison Test for Series). [cor:comparison-test] *Let F be a complete ordered field, $a, b : \mathbb{N}_1 \to F$ with $0 \leq a_n \leq b_n$ for all $n \in \mathbb{N}_1$. If the series of b is convergent, then the series of a is convergent and $\sum a \leq \sum b$. Follows by applying theorem monotone comparison increasing to the partial sum sequences* $\text{series}(a)$ *and* $\text{series}(b)$*, both of which are increasing by lemma 5.1.*

Theorem 5.1 (Absolute Convergence Implies Convergence). [thm:absolute-convergence] *Let F be a complete ordered field and $a : \mathbb{N}_1 \to F$. If the series of $|a|$ (where $|a|(n) := |a(n)|$) is convergent, then the series of a is convergent.*

Proof. **TODO** □

Remark 5.1 (Two Methods for Checking Series Convergence). [rem:two-methods-series] Depending on the sign structure of the terms of a:

- *Single-sign* (all $a(n) \geq 0$ or all $a(n) \leq 0$): apply the Comparison Test for Series directly. Monotonicity of $\text{series}(a)$ comes from lemma 5.1 (or the sign-flipped version).
- *Mixed-sign*: apply Absolute Convergence first, reducing the question to $\text{series}(|a|)$ (non-negative), where the Comparison Test applies.

6 Special Series

Theorem 6.1 (Harmonic Series Diverges)**.** [thm:harmonic-series-diverges] *The series of the harmonic sequence $a(n) := 1/n$ is divergent.*

Proof. **TODO** □

Theorem 6.2 (Reciprocal Squares Series Converges)**.** [thm:reciprocal-squares-converges] *The series of $a(n) := 1/n^2$ is convergent.*

Proof. **TODO** □